Course Name: COMP5334-Advanced Network and Computer Security

Semester: Summer II – 2026

Instructor: Dr. Azamat Zhamanov

Project Title: Enterprise Network Design and Security Using Cisco Packet Tracer

Submitted By: Yusra Jalileh

Date: July 31, 2026

Introduction

Modern business relies on computers for most of its day-to-day operations. These computer networks enable employees to communicate with one another and with other devices and to share information.

Therefore, it is vital to implement effective security mechanisms for all computer networks to ensure their protection and the reliability of communications. This project aimed at designing and configuring an enterprise level secure network environment using Cisco Packet Tracer. The network was divided into multiple subnetworks based on VLSM. Each subnetwork was connected via a router and switches. Various network services were established to provide connectivity among devices in a secure manner. Technologies such as DHCP (for assigning IP addresses automatically), inter-network routing (for connecting subnetworks), switch management (for managing switches), port security (for controlling access to ports on switches), and Access Control Lists (ACLs) (for restricting communication between certain networks) were utilized throughout the course of this project.

Additionally, connectivity tests were conducted to validate the functionality of the configured security policies and the overall operation of the network. The subsequent sections of this report detail the design of the network, the IP addressing plan used in the network, how the network was configured, how the security features of the network were implemented, and the results of the connectivity tests conducted to demonstrate the successful operation of the network.

2. Network Topology

The above network has been configured using Cisco Packet Tracer. It includes a router, three switches, one wireless access point, three PCs, two laptops, and three servers. The router links all the subnetworks for communication purposes.

The first subnetwork has the wired PCs linked through Switch0. The second subnetwork has the wireless subnetwork, where laptops are linked through the access point and Switch1. The third subnetwork has the servers connected through Switch2. This is a way of segmenting the network into various parts for management and security purposes.

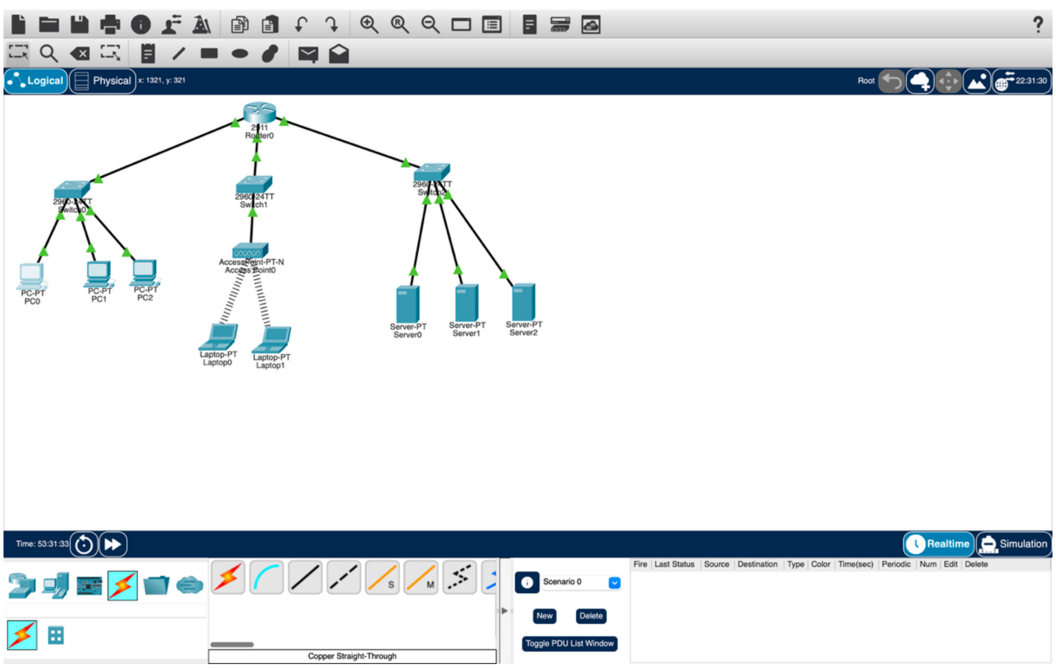


Figure 1. Network Topology

3. Network Addressing

The network was split up into three distinct subnets using VLSM. Different masks were used for each of the subnets according to the number of hosts needed. The router was configured as the default gateway for each of the subnets.

Network Addressing Table

Network Address	Broadcast Address	Subnet Mask	Wildcard Mask
192.168.0.0	192.168.0.63	255.255.255.192 (/26)	0.0.0.63
192.168.0.64	192.168.0.95	255.255.255.224 (/27)	0.0.0.31
192.168.0.96	192.168.0.111	255.255.255.240 (/28)	0.0.0.15

Device Addressing Table

Device	Interface	IP Address	Subnet Mask
Router	GigabitEthernet0/0	192.168.0.1	255.255.255.192
Router	GigabitEthernet0/1	192.168.0.65	255.255.255.224
Router	GigabitEthernet0/2	192.168.0.97	255.255.255.240
Switch0	VLAN1	192.168.0.10	255.255.255.192
Switch1	VLAN1	192.168.0.66	255.255.255.224
Switch2	VLAN1	192.168.0.101	255.255.255.240
PC0	FastEthernet0	192.168.0.2	255.255.255.192
PC1	FastEthernet0	192.168.0.3	255.255.255.192

Device	Interface	IP Address	Subnet Mask
PC2	FastEthernet0	192.168.0.4	255.255.255.192
Laptop0	Wireless0	192.168.0.68	255.255.255.224
Laptop1	Wireless0	192.168.0.67	255.255.255.224
DHCP Server	FastEthernet0	192.168.0.98	255.255.255.240
Web Server	FastEthernet0	192.168.0.99	255.255.255.240
NTP Server	FastEthernet0	192.168.0.100	255.255.255.240

The gateway addresses for each router interface include:

* GigabitEthernet0/0 – 192.168.0.1

* GigabitEthernet0/1 – 192.168.0.65

* GigabitEthernet0/2 – 192.168.0.97

The gateway addresses help devices in each of the subnets to be able to connect to other networks using the router.

4. DHCP Configuration

Dynamic Host Configuration Protocol (DHCP) was configured for the automatic assignment of IP addresses in the wireless subnet. This reduces the need for manual configuration of IP addresses. The DHCP server was configured with the appropriate address pool, subnet mask, default gateway, and the maximum number of addresses. Once the DHCP was configured properly, the laptops obtained IP addresses automatically.

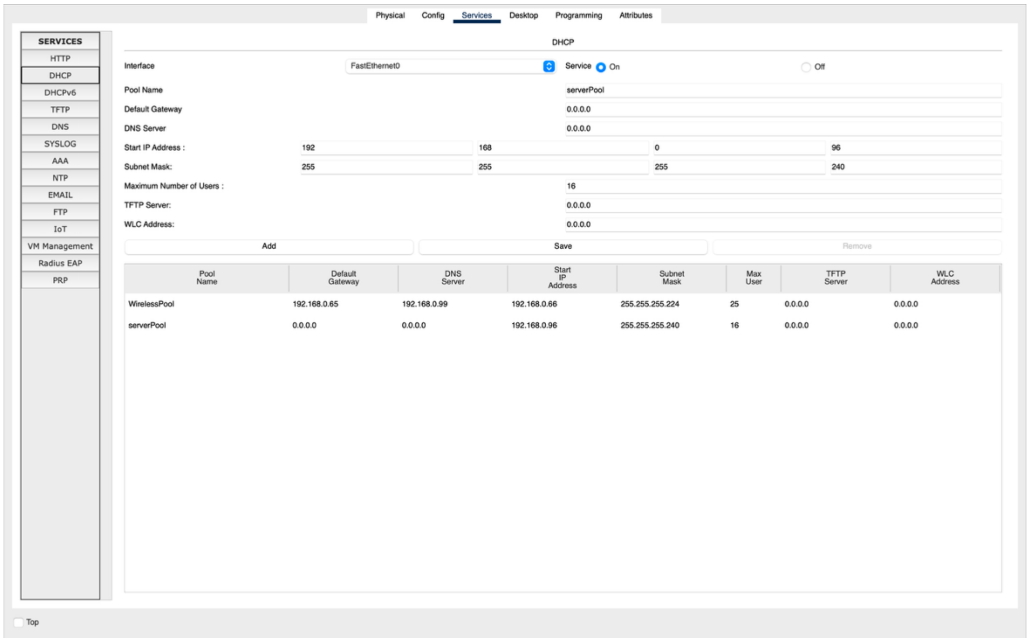


Figure 2. DHCP Configuration

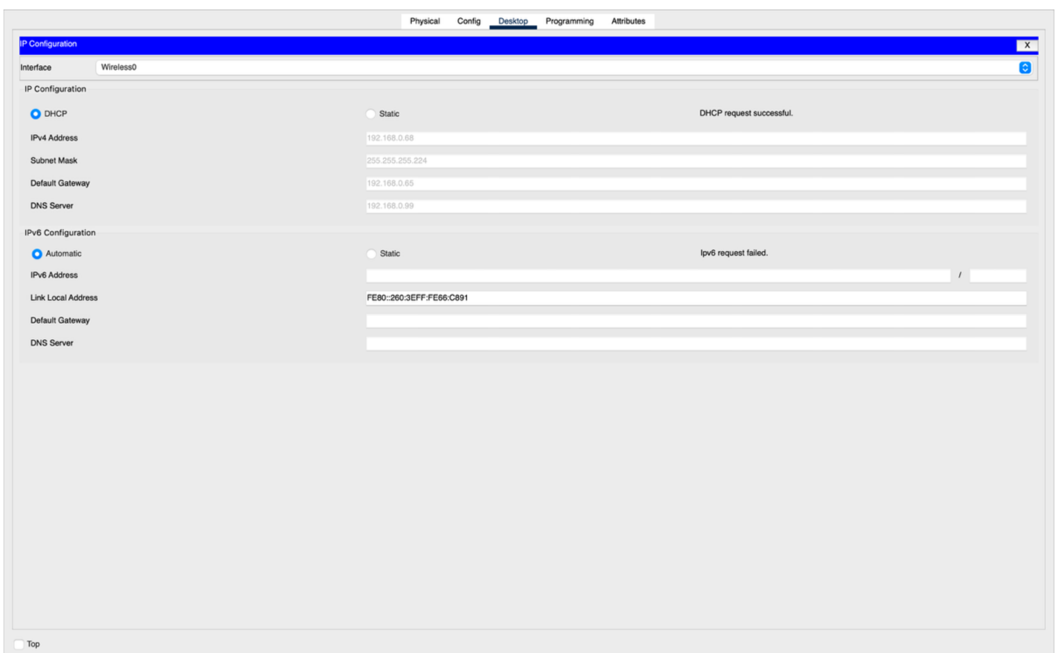


Figure 3. Laptop DHCP Lease

5. Switch Management

Each switch was configured with a management IP address using VLAN 1. Also, the default gateway was configured for the purposes of managing the switches from various subnetworks.

The management IP addresses of the switches in this project are as follows:

* Switch0 -> 192.168.0.10

* Switch1 -> 192.168.0.66

* Switch2 -> 192.168.0.101

Once the above configuration is done, the switches were able to communicate with the router.

6. Port Security Configuration

Port Security was enabled on Switch0 with the aim of increasing the security of the network by restricting the number of devices connecting to a particular switch port. This has been achieved by configuring Port Security on interface FastEthernet0/2, which is connected to PC0.

The port has been configured for access, and it supports learning of one MAC address through the sticky configuration. Violation mode is restricted, which means that the port remains operational but unauthorized traffic is blocked.

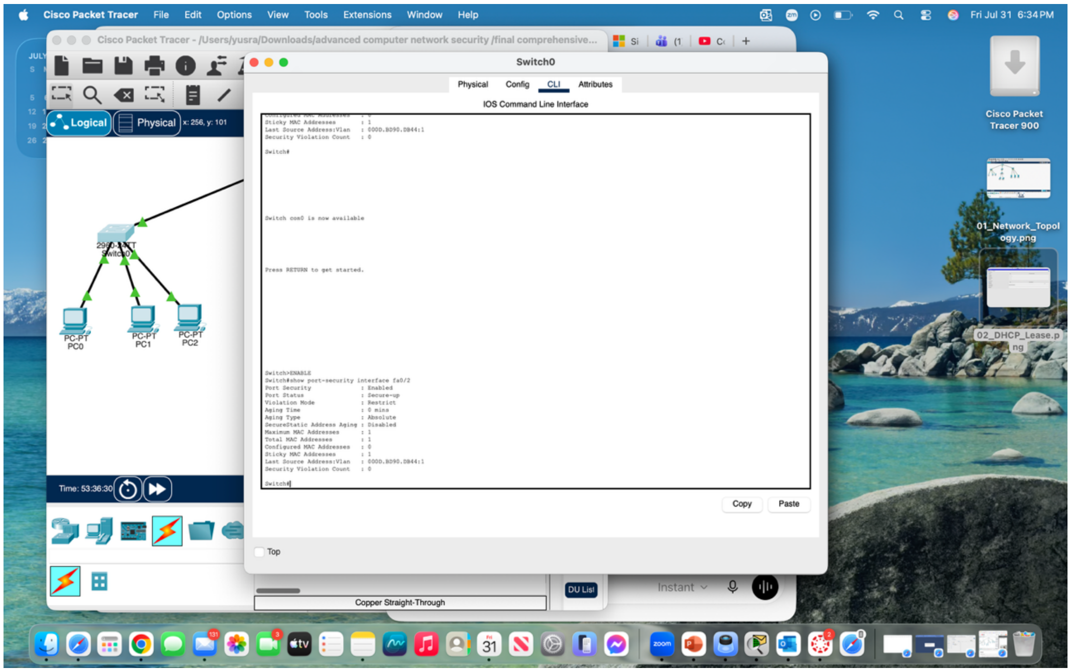


Figure 4. Port Security Configuration

7. Extended Access Control List (ACL) Configuration

Extended ACL 101 was created on the router in order to regulate communication between different subnets. The ACL was applied in the inbound direction on interface GigabitEthernet0/0.

The security policy in place provides the ability for PC0 to reach server network but prevents communication with the wireless network. All other communication is allowed in order to allow normal communication for other devices.

Below there is the configuration of ACL:

* Deny communication of PC0 (192.168.0.2) to wireless network 192.168.0.64/27.

* Allow communication of PC0 to the server network 192.168.0.96/28.

* Allow all other IP communications.

This example illustrates how extended ACL can be used to create basic security policies in the enterprise environment.

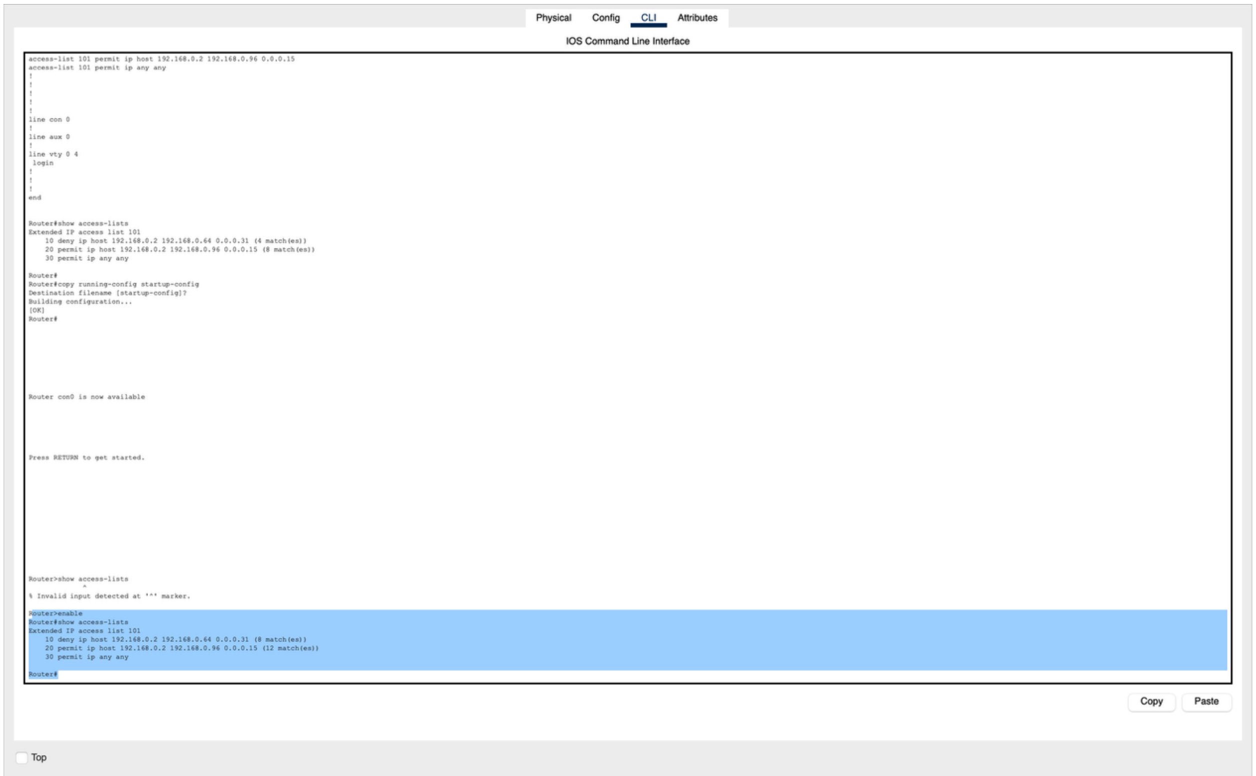


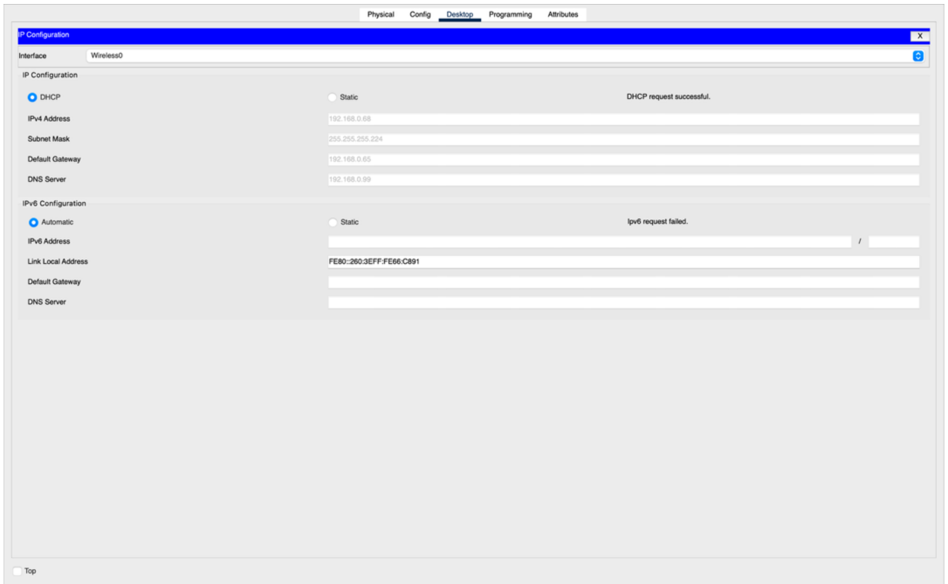
Figure 5. ACL Verification

8. Testing and Verification

Various tests were performed after configuring the network in order to make sure all services worked properly.

a. DHCP Test

The wireless laptop obtained an IP address automatically from the DHCP server. All received values such as IP address, subnet mask, default gateway, and DNS server prove that the DHCP service works properly.



(Figure 3 – Laptop DHCP Lease)

b. ACL Test

A successful ping test was performed from PC0 to Server1 (192.168.0.99). This proves that ACL permits access to the server network.

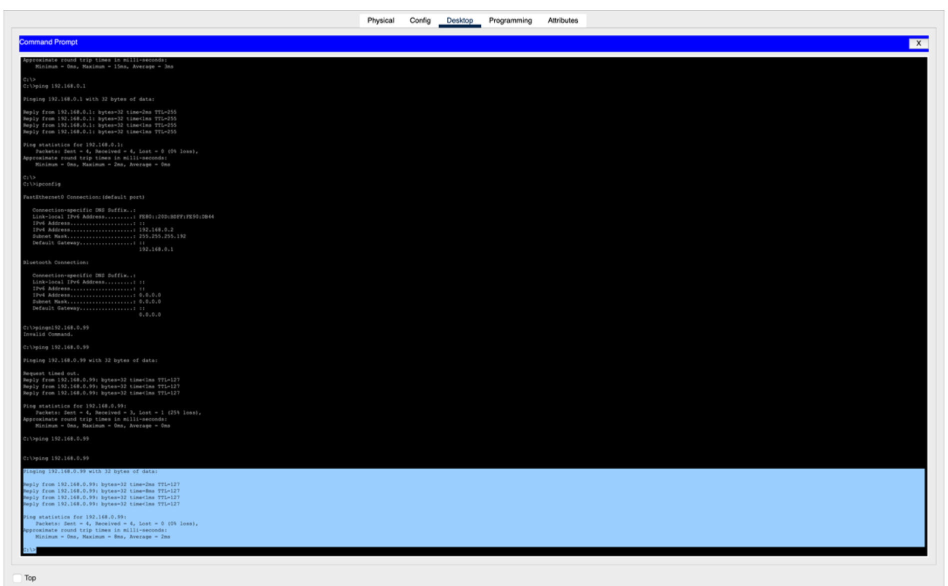


Figure 6. Successful Ping (ACL Permit)

Then another ping test was performed from PC0 to Laptop0 (192.168.0.68). This ping failed because ACL 101 blocks communication from PC0 to the wireless subnet.

.

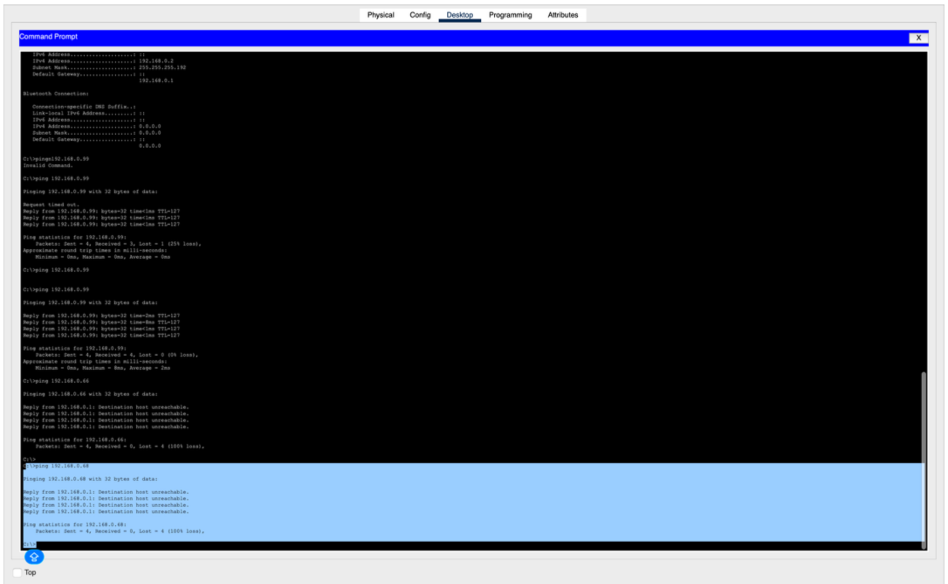


Figure 7. Blocked Ping (ACL Deny)

c. Routing Verification

The router routing table was viewed using the "show ip route" command. The output proves that all three subnetworks were directly connected to the router via proper interfaces.

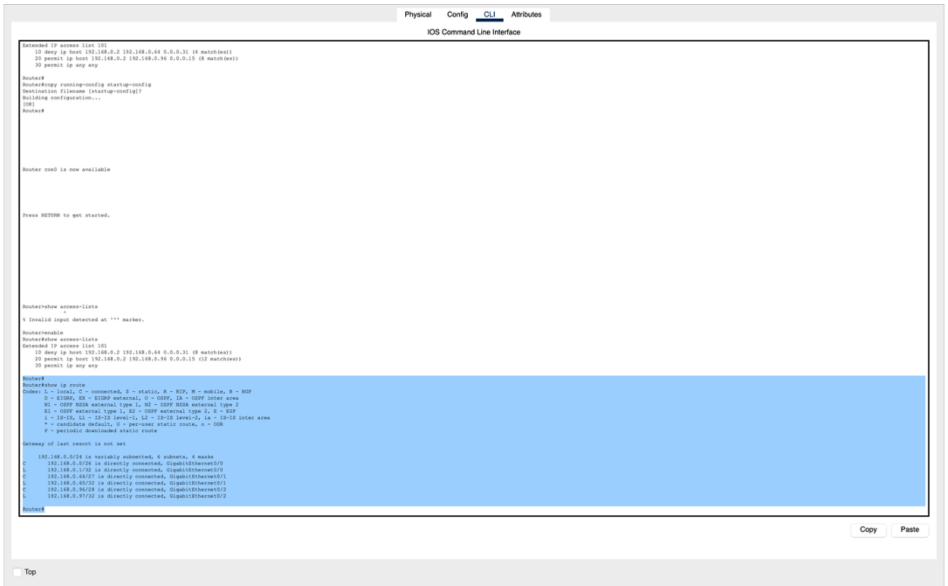


Figure 8. Routing Table

d. Verification of Interface

The "show IP interface brief" command was issued to verify the status of interfaces of the router. It can be seen from the output that all configured interfaces are in the "up/up" state.

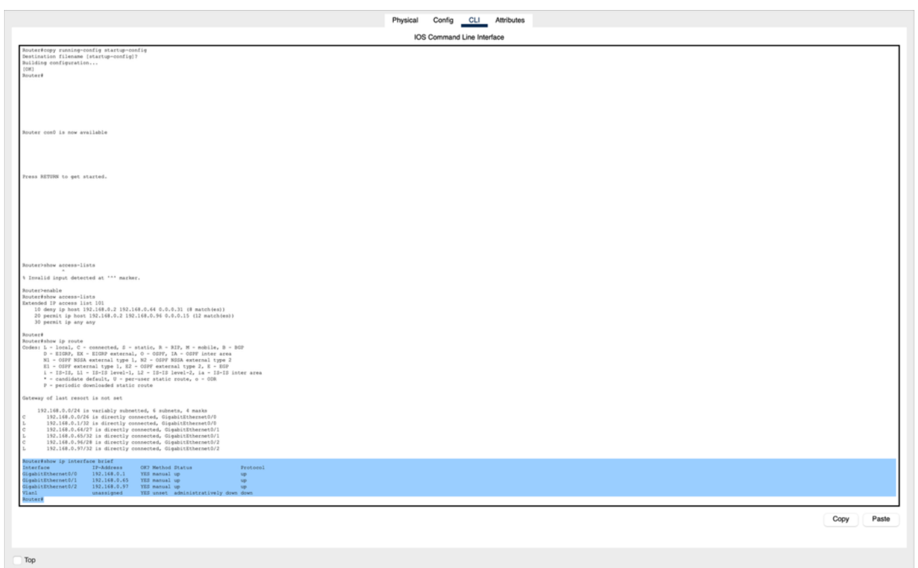


Figure 9. Router Interfaces

9. Conclusion

In this project, I was able to apply what I have learned in configuring and securing a small enterprise network in Cisco Packet Tracer. A number of networking concepts such as subnetting, routing, DHCP, switch configuration, port security, and ACLs were applied.

The completed network enabled the interconnection between all subnets and ensured proper security policy enforcement using ACLs and Port Security. It was found out that the network services were working well and security measures were working properly.

Generally, this project improved my knowledge on the design of enterprise networks.